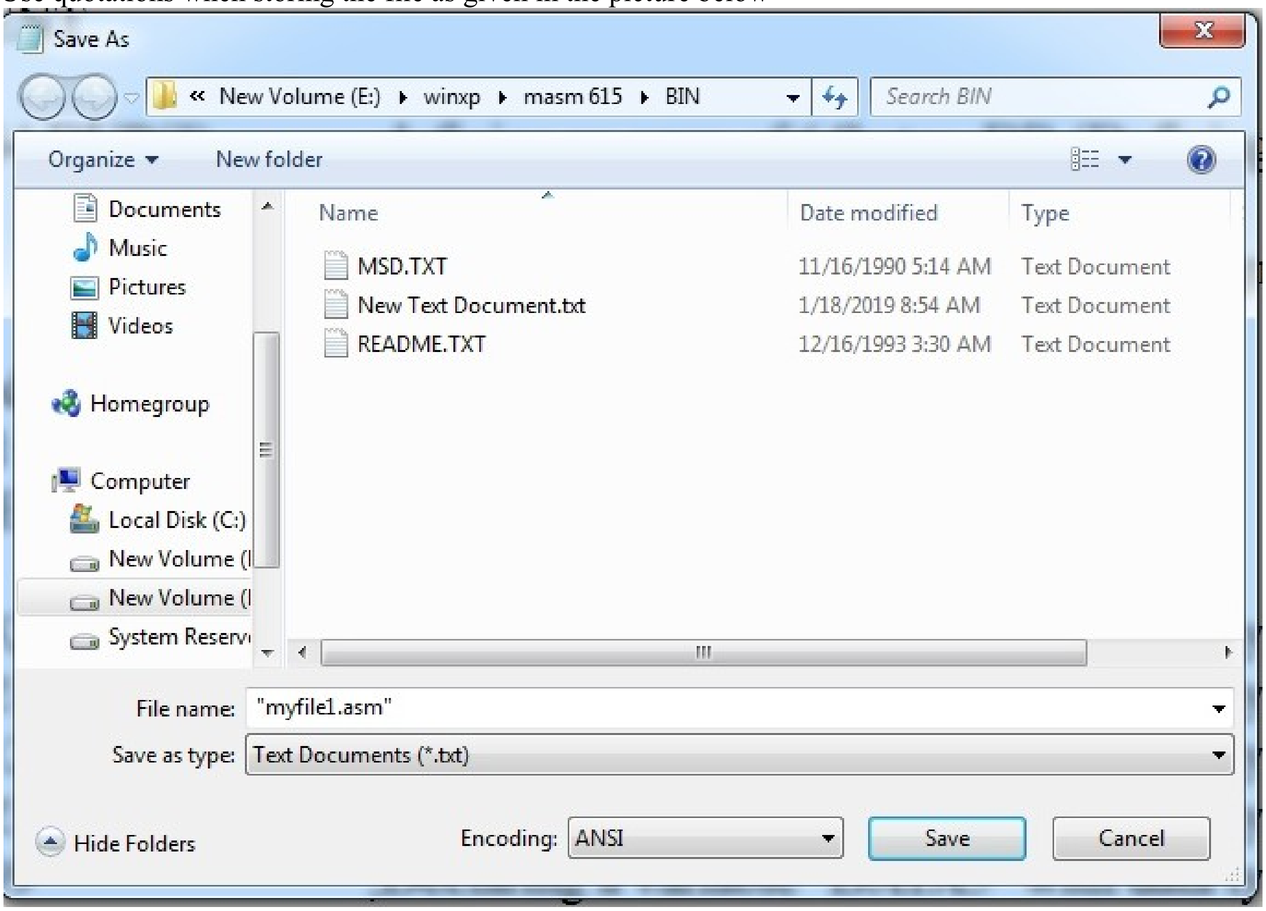
PROGRAM

```
;Segment defined as stack segment and named as "STSEG"
;It can be named with programmer's choice but must not be a keyword
STSEG SEGMENT
        DB 64 DUP(?)  ;defining an array of 64bytes, DB (Defining Byte) and DUP (Duplicate)
STSEG ENDS            ;stack segment ends here, ENDS (End Segment)

;Segment defined as data segment and named as "DTSEG"
;It can be named with programmer's choice but must not be a keyword
DTSEG SEGMENT

DATA1       DB 32  ;Declaring a variable "DATA1" with data type byte (DB) and initializing with value 32 decimal
DATA2       DB 21  ;Declaring a variable "DATA2" with data type byte (DB) and initializing with value 21 decimal
DATA3       DB 12  ;Declaring a variable "DATA3" with data type byte (DB) and initializing with value 12 decimal
DATA4       DB 23  ;Declaring a variable "DATA4" with data type byte (DB) and initializing with value 23 decimal
DATA5       DB 39  ;Declaring a variable "DATA5" with data type byte (DB) and initializing with value 39 decimal
SUM         DB 0   ;Declaring a variable "SUM" with data type byte (DB) and initializing with value 0 decimal

DTSEG ENDS         ;data segment ends here, ENDS (End Segment)

;Segment defined as code segment and named as "CDSEG"
;It can be named with programmer's choice but must not be a keyword
CDSEG SEGMENT

MAIN PROC          ;main procedure starts here (similar to void main (void))

;informing the code about assumptions that STSEG is SS, DTSEG is DS, and ;CDSEG is CS using
;ASSUME directive
ASSUME SS:STSEG, DS:DTSEG, CS:CDSEG

MOV AX, DTSEG          ;initializing the data segment by moving the value of DTSEG in DS
MOV DS, AX

MOV AL, DATA1          ;AL = 32
ADD AL, DATA2          ;AL = AL + DATA2
ADD AL, DATA3          ;AL = AL + DATA3
ADD AL, DATA4          ;AL = AL + DATA4
ADD AL, DATA5          ;AL = AL + DATA5

MOV SUM, AL            ;SUM = AL

;transferring the control to the operating system by using interrupt 21h
MOV AH, 4CH        ;AH = 4C (hexadecimal)
INT 21H

MAIN ENDP          ;main procedures ends here, ENDP (End Procedure)

CDSEG ENDS         ;code segment ends here, ENDS (End Segment)

END MAIN           ;the whole code (program) ends here
```

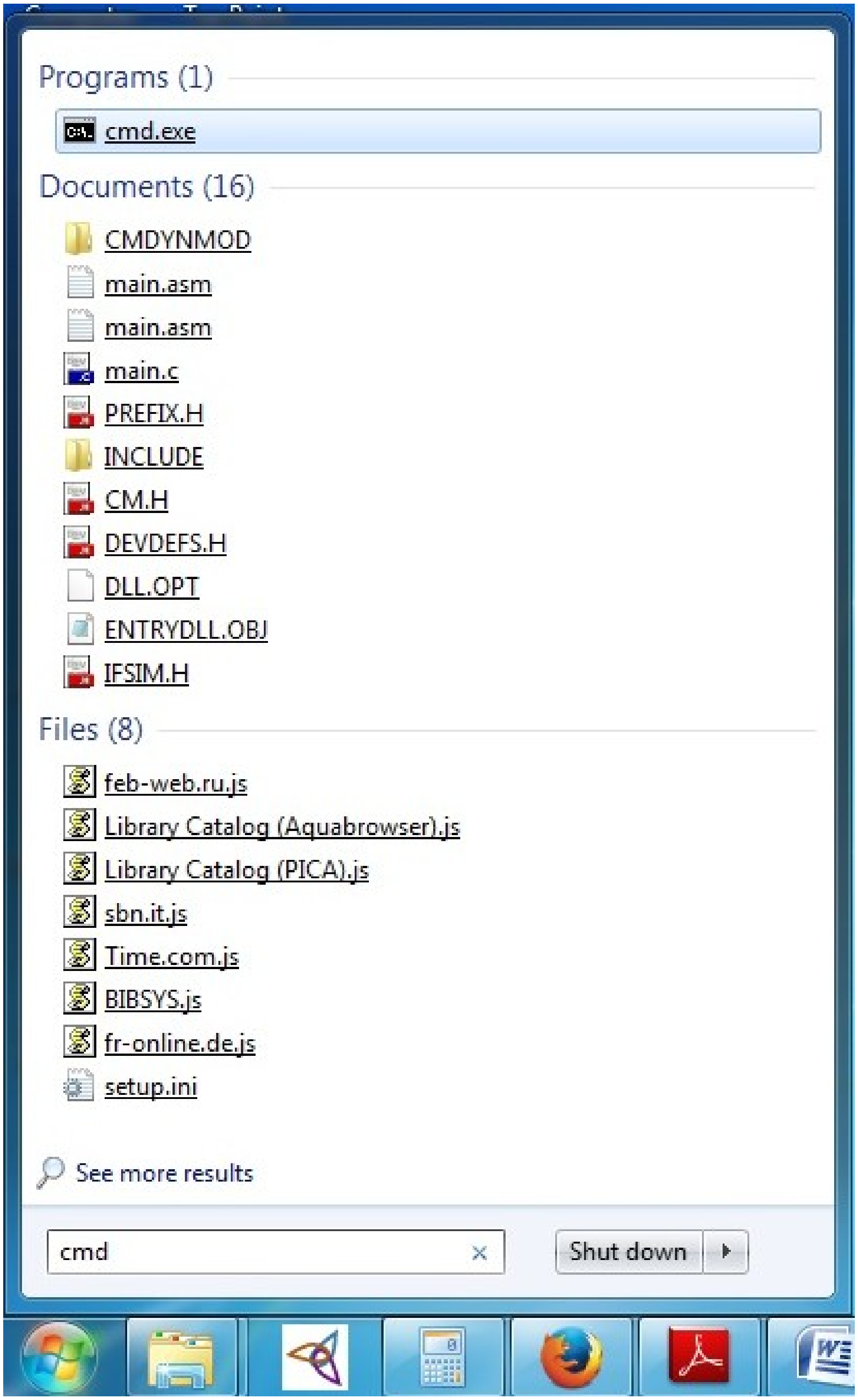
Assembling and Linking the program to create Executable file
Download and extract MASM615 zip file in drive E (as per your convenient).
In the same drive E, copy **winxp** folder.

Write the program using NOTEPAD or any other EDITOR software
Store it as myfile.asm, in the BIN folder under MASM615.

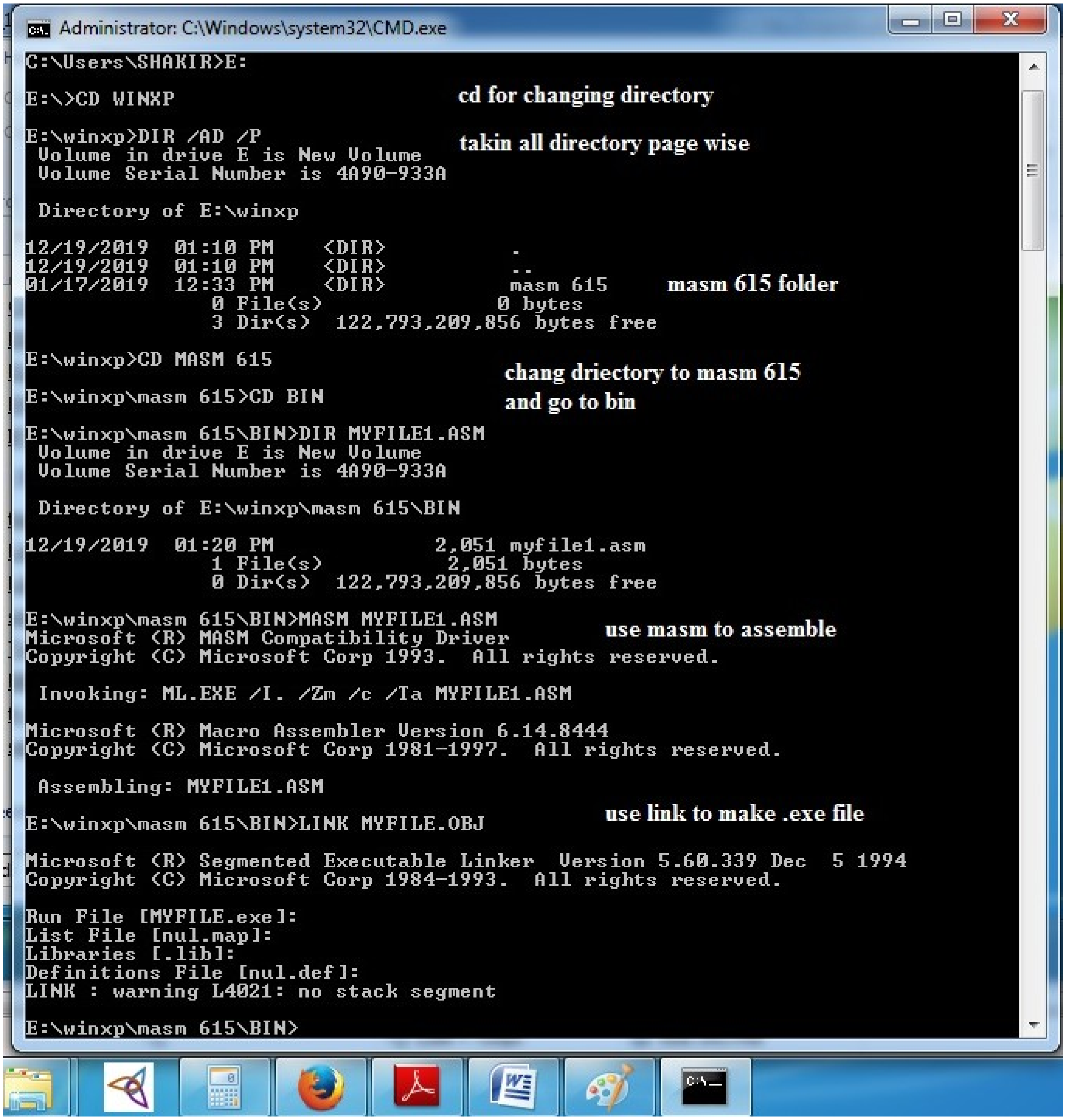
Use quotations when storing the file as given in the picture below



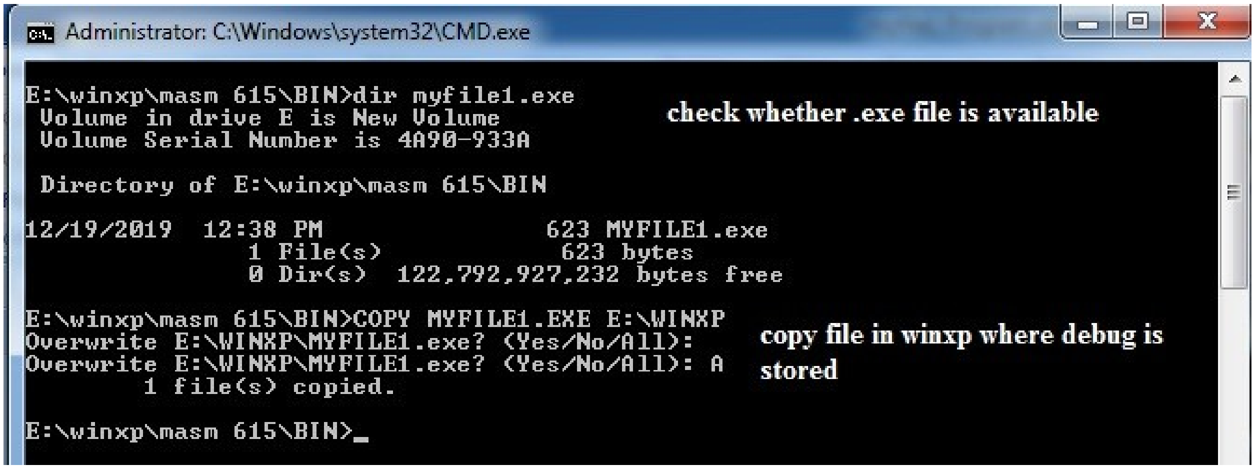
Go to command line window.



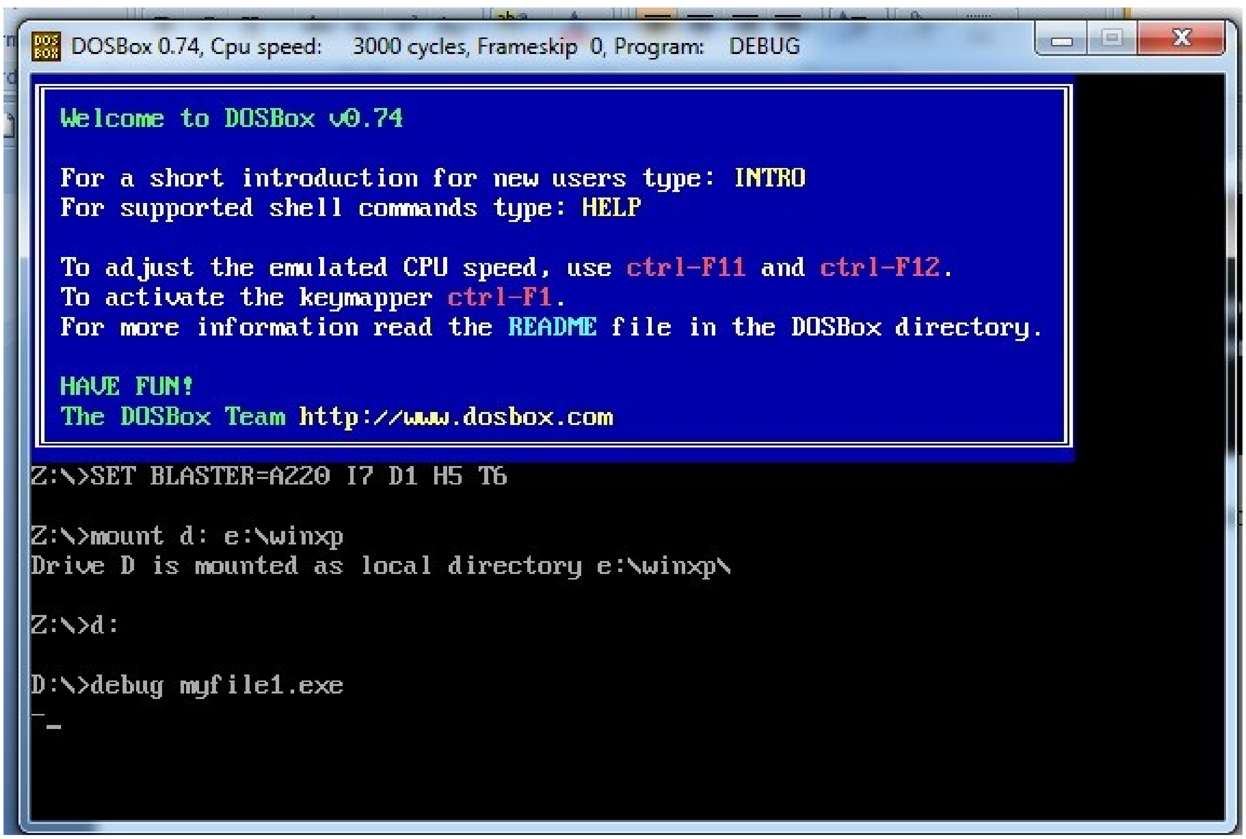
cmd window appears. Do as describe in the following figure.



Copy the .exe file just created in the folder where debug utility is stored. It is described in the following figure.

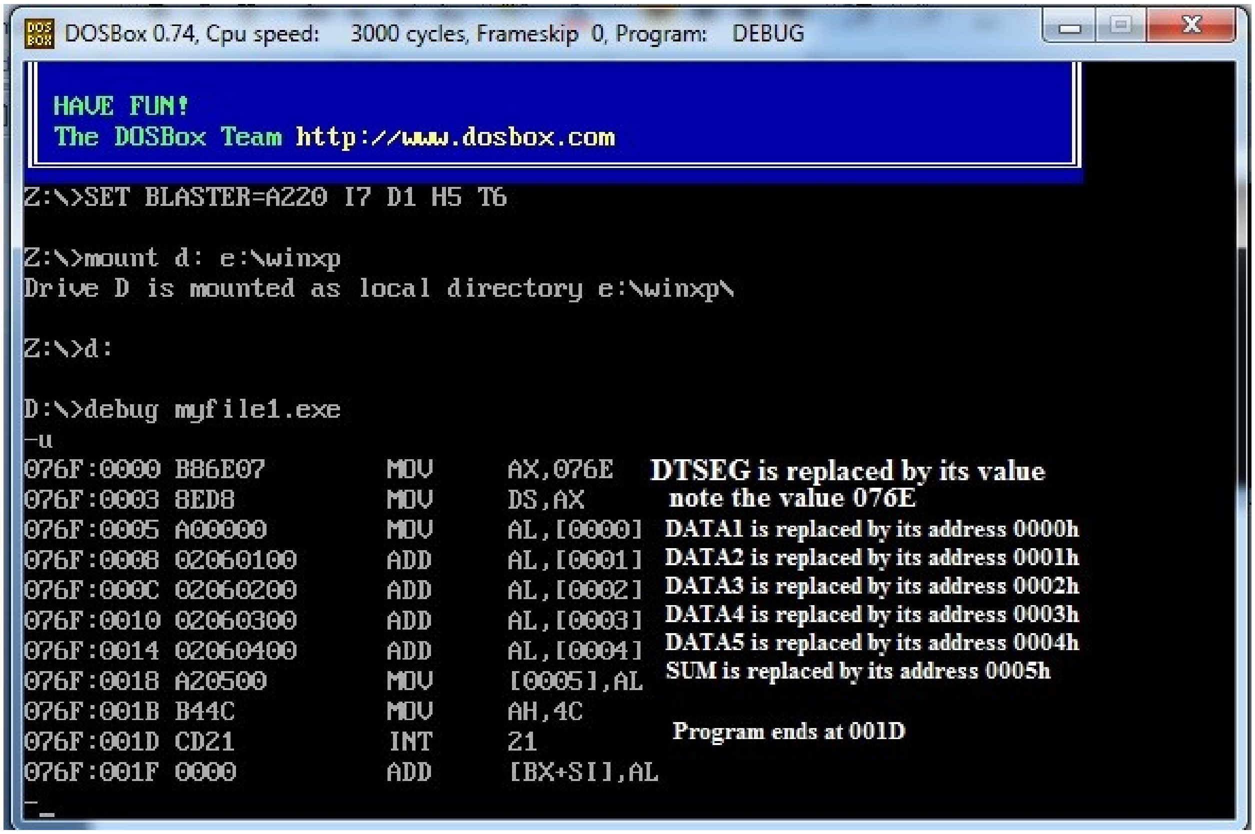


Now, open DOSBox, mount and open the .exe file in debug. The process is shown in the figure below.



Use the exetension ".exe" is mandatory.

Further, use unassembled command to analyze the program.

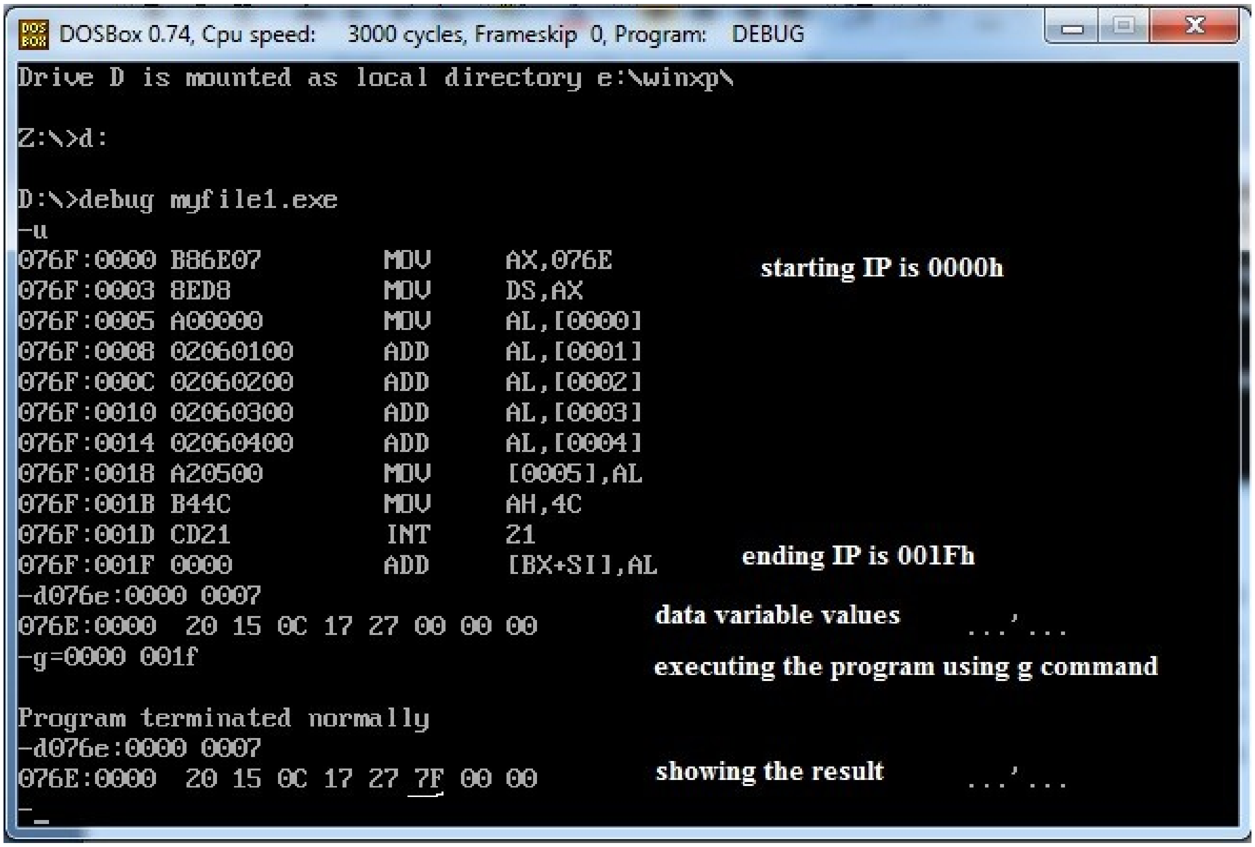


Data Segment value is 076E.
The variable DATA1 is located on the memory location 076E:0000
The variable DATA2 is located on the memory location 076E:0001
The variable DATA3 is located on the memory location 076E:0002
The variable DATA4 is located on the memory location 076E:0003
The variable DATA5 is located on the memory location 076E:0004
The variable SUM is located on the memory location 076E:0005

To check the data values stored in data segment write the command

- D 076e:0000 0007

As discussed in the following figure.



The sum is 7Fh.